SUMS Proofs Workshop: Fall

Note: Problems marked with (*) are difficult. (**) indicates *very difficult*.

Starter Problems

Problem 1 Prove that 0 divides an integer a if and only if $a = 0$.

Problem 2 Prove that there do not exist integers m and n such that $14m + 21n = 100$.

Problem 3 Prove that n^2 odd implies that n is odd.

Problem 4 Show that $\sqrt{2}$ is irrational.

Problem 5 Show that $1^2 + 2^2 + 3^2 + \cdots + n^2 = \frac{n(n+1)(2n+1)}{6}$ for every positive integer n.

Problem 6 For all $n \in \mathbb{N}$ show that $2^n \leq 3^n - 1$.

Problem 7 Let $n \in \mathbb{Z}$. Suppose there exists $a \in \mathbb{Z}$ such that $n = a^2 - a$. Show that n is even.

Problem 8 Let $f : \mathbb{R} \to \mathbb{R}$ be $f(x) = 3x^2 + 2$. Show that this is not injective.

Problem 9 Let $\sim$ be an equivalence relation on a set X. Prove that for all $a, b \in X$, $a \sim b$ if and only if they have equal equivalence classes.

Problem 10 Let $\sim$ be an equivalence relation on a set X. Prove that for all $a, b \in X$,

$$[a]_\sim = [b]_\sim \text{ or } [a]_\sim \cap [b]_\sim = \varnothing.$$

You may use the fact $[a]_\sim = [b]_\sim$ iff $a \sim b$ without proof (this is the previous problem!).

Problem 11 Prove that

$$1 \cdot 1! + 2 \cdot 2! + 3 \cdot 3! + \cdots n \cdot n! = (n+1)! - 1.$$

Note $n! = n \cdot (n-1) \cdot (n-2) \cdot 2 \cdot 1$.

Problem 12 Define a relation $\sim$ on $\mathbb{N} \times \mathbb{N}$, where $0 \notin \mathbb{N}$, and

$$(a, b) \sim (c, d) \iff ad = bc.$$

Prove that $\sim$ is an equivalence relation.

Problem 13 Prove that if $|a + b| > 2$, then $|a| > 1$ or $|b| > 1$, where $a, b \in \mathbb{R}$.

Problem 14 Your friend claims that

$$1 + 2 + \cdots + n = \frac{n(n+3)}{2} - 1.$$

The base case $n = 1$ holds. Does the claim hold in general? If so prove it, if not disprove it.

PROBLEM 15 Suppose $a, b, c \in \mathbb{Z}$. Prove $a^2 + b^2 = c^2$ implies either a or b is even.

PROBLEM 16 Suppose k, n are natural numbers, excluding 0. Then $k \nmid n$ if and only if there exist $q, r \in \mathbb{Z}$ with $0 < r < |k|$ such that $n = qk + r$. Hint: Use the division theorem: For any $n, k \in \mathbb{N}$, there exists unique $q, r \in \mathbb{N}$ such that $n = qk + r$ and $0 \leq r < |k|$.

PROBLEM 17 If $n, m \in \mathbb{N}$, prove that if $m|n$ and $n|m$ then $n = m$.

PROBLEM 18 For sets A, B, C show that $A \cap (B - C) = (A \cap B) - (A \cap C)$. Is it always true thar $A \cup (B - C) = (A \cup B) - (A \cup C)$?

PROBLEM 19 (From Berkeley's Intro to Proofs) Prove that the composition of any two decreasing functions is increasing. Hint: Given decreasing functions f and g, and numbers $x_1 < x_2$, show that $f(g(x_1)) < f(g(x_2))$.

PROBLEM 20 Prove that if x and y are real numbers, then $\max(x, y) + \min(x, y) = x + y$.

Harder Problems

PROBLEM 21 (*) Assume that the following statement is true: if Micah dislikes pineapple pizza then he likes pineapple pizza. Does he like pineapple pizza?

PROBLEM 22 (*) For functions $f : X \to Y$ and $g : Y \to X$ with $g \circ f = \mathrm{Id}_X$, decide the truth of:

1. f is surjective.
2. f is injective.
3. g is surjective.
4. g is injective.

PROBLEM 23 (*) Show that $\mathbb{N}$ has the same size as $\mathbb{Z}^+$ (the positive integers).

PROBLEM 24 (*) Negate the following definitions without using the word "not" or the symbol $\neg$:

1. $\forall x \in \mathbb{R} \setminus \mathbb{Q}$, if $x^2 \in S$ then $x^3 \in T$.
2. For all $x, y \in \mathbb{R}$, if $x, y \in \mathbb{Q}$ then $x + y \in \mathbb{Q}$.
3. We say $f : X \to Y$ is continuous at $p \in X$ if $\forall \epsilon > 0$, $\exists \delta > 0$ s.t. $\forall x \in E$ with $d_X(x, p) < \delta$, then $d_Y(f(x), f(p)) < \epsilon$ (†).
4. For some $f : X \to Y$, we say $\lim_{x \to p} f(x) = L$ if $\forall \epsilon$, $\exists \delta > 0$ s.t. $\forall x \in X$ with $d_X(x, p) < \delta$, then $d_Y(f(x), L) < \epsilon$ (†).

PROBLEM 25 (*) Consider the claim "$\mathcal{P}(A \cup B) \subseteq \mathcal{P}(A) \cup \mathcal{P}(B)$." If true, prove it. Otherwise, disprove it. Note that $\mathcal{P}(S)$ indicates the powerset of S.

Problem 26 (*) Define a sequence of numbers by $a_0 = 1$ and $a_{n+1} = 3a_n + 2$. Prove $a_n = 2 \cdot 3^n + 1$.

Problem 27 (*) For sets A, B, C, is $(A \setminus B) \cup (B \setminus C) \subseteq A \setminus C$? If true prove it, else disprove it.

Problem 28 (*) Prove or give a counterexample: $(X \cap Y) \times (Z \cap W) = (X \times Z) \cap (Y \times W)$, for any sets X, Y, Z, W.

Problem 29 (*) Show that $a \sim b$ with $a, b, \in \mathbb{N} \iff$ the sum of the decimal digits of a and b are congruent mod 9 is an equivalence relation.

Problem 30 (*) Let $S = \{0, 1\}^n$ be the set of n-digit binary strings. Define $h(x) =$ the number of 1s present in x, for $x \in S$ to be the *hamming weight* of x. Prove that $a \sim b \iff h(a) = h(b)$, for $a, b \in S$, is an equivalence relation.

Challenging Problems

Problem 31 (**) Show that

$$\bigcap_{n=1}^{\infty} \left(\frac{-1}{n}, \frac{1}{n} \right) = \{0\}.$$

Note: You may use without proof that, given any real number, there is another real number greater than it.

Problem 32 (**) The *Fibonacci sequence* is defined by $F_0 = 0$, $F_1 = 1$, and

$$F_n = F_{n-1} + F_{n-2}$$

for $n \geq 2$. Prove the following identities for all $n \geq 0$:

$$F_{2n} = F_n(F_{n+1} - F_{n-1}) \quad F_{2n+1} = F_{n+1}^2 + F_n^2$$

Hint: Use strong induction on both identities simultaneously.

Problem 33 (**) The *Fibonacci sequence* is defined by $F_0 = 0$, $F_1 = 1$, and

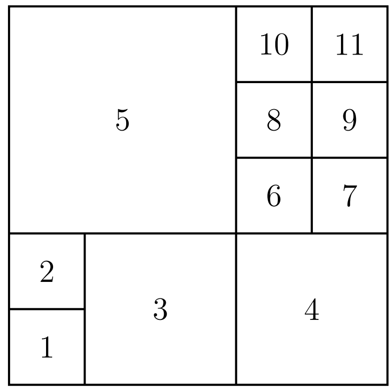
$$F_n = F_{n-1} + F_{n-2}$$

for $n \geq 2$. Show that every positive integer can be written as the sum of distinct Fibonacci numbers. Distinct means the sum cannot have repeated values; For example, $2 = 1 + 1$ is not a valid sum even though $F_1 = 1$ and $F_2 = 1$. But $2 = 2 = F_3$ is.

Problem 34 (**) (From the 2002 Putnam, A2). Given any five points on a sphere, show that some four of them lie in a closed hemisphere.

Problem 35 (**) Given 51 distinct positive integers strictly less than 100, prove two of them sum to 99.

Problem 36 (**) (From this NYT Puzzle) A positive integer n is called *squareable* if we can build a square out of n smaller squares, not necessarily of the same size. For example, 11 is squareable:



Prove that all integers greater than 5 are squareable. Hint: Prove by induction with three base cases.

Problem 37 (**) At a party, prove that the number of people who shook hands an odd number of times is even.

Problem 38 (**) There exist two positive irrational numbers s and t such that s^t is rational. Hint: Take $s = t = \sqrt{2}$. Prove by cases.

Problem 39 (**) For any positive integer d, there exists a number composed only of the digits 7 and 0 such that d divides this number.

Problem 40 (**) What is the minimum number of breaks to break up a chocolate bar made of $m \times n$ little pieces into those very pieces?